PROVISIONAL PATENT APPLICATION SPECIFICATION

TITLE OF THE INVENTION

Nasal Hemostatic Device with Integrated Vasoconstrictor Delivery and Absorbent Deployment

FIELD OF THE INVENTION

The present invention relates generally to medical devices for the management of nasal hemorrhage (epistaxis), and more particularly to a reusable handheld device that, in a single actuation, drives an absorbent element through a sealed cartridge of vasoconstricting liquid and into the nasal cavity, simultaneously delivering the vasoconstricting agent to the bleeding site and absorbing blood.

BACKGROUND OF THE INVENTION

Nosebleeds, or epistaxis, are among the most common minor medical events. Approximately 60 percent of individuals experience a nosebleed at some point in their lives, and approximately 15 percent experience them regularly. Nosebleeds arise from many causes, including direct impact to the nose, a scratched or damaged blood vessel within the nasal cavity, and dry or changing climate conditions.

Existing methods for stopping a nosebleed suffer from significant drawbacks. Pinching the nostrils requires sustained manual pressure for extended periods and causes soreness. Packing the nostril with tissue or gauze is messy, difficult to position correctly, and frequently allows blood to drain down the throat. Topical vasoconstrictor sprays require the user to separately apply the agent and then manage absorption. Many existing products perform inconsistently across different users and situations, and in severe cases bleeding may continue for as long as approximately two hours. Existing solutions are slow, messy, awkward to use, and frequently inaccessible at the moment a nosebleed occurs, which is particularly problematic for athletes who must leave play until bleeding stops.

A need therefore exists for a device that places both a vasoconstricting agent and an absorbent element precisely at the bleeding site in a single intuitive motion, that is compact enough to store in a first aid kit, nurse's office, or sports bag, that is reusable and economical, and that functions reliably across users of different ages and across different situations.

SUMMARY OF THE INVENTION

The invention is a reusable handheld nasal hemostatic device. In a single push action resembling the operation of a syringe, the device drives an absorbent cotton element upward through the interior of a sealed replaceable cartridge containing a vasoconstricting liquid. The cotton element punches through the cartridge's frangible seals, absorbs the vasoconstricting liquid as it passes through the cartridge interior, and exits through a soft nose head into the user's nostril. The deployed cotton element thereby simultaneously delivers the vasoconstricting agent to the bleeding blood vessel and absorbs blood.

The device comprises five principal components: (1) an elongated tubular main body having a finger-grip flange at its lower end; (2) a stem received through the lower end of the main body via a keyed interface, the stem and body cooperating to provide a twist-to-lock priming mechanism and a full-stroke deployment travel; (3) a replaceable cylindrical cartridge, sealed at top and bottom and containing a vasoconstricting liquid, seated in the upper bore of the main body; (4) a replaceable absorbent cotton element loaded into the lower bore of the main body; and (5) a flexible nose head that threads onto the upper end of the main body, retains the cartridge in place, and presents a rounded dome for atraumatic insertion into the nostril.

Only the cartridge and the cotton element are consumed in use. The main body, stem, and nose head are reusable indefinitely. The cartridge is a low-cost disposable component, and the cotton element is biodegradable, together reducing waste relative to single-use nasal packing products and keeping the device economical for users who experience frequent nosebleeds.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a perspective view of the assembled device.

FIG. 2 is an exploded perspective view of the device, showing the stem, the absorbent cotton element, the main body, the cartridge, and the nose head arranged along a common axis in their order of assembly.

FIG. 3 is a perspective detail view of the lower end of the main body, showing the internal locking collar with the locking pocket, the deployment channel, the guide walls, and the transverse turning region.

FIG. 3A is a view of the lower face of the base flange of the main body, as seen from below, showing the lower opening, the keyed receiving feature, and the directional markings.

FIG. 4 is a perspective view of the main body inverted, showing the lower face, the lower opening, the keyed receiving feature, and the locking pocket and deployment channel of the internal collar.

FIG. 5 is a perspective cutaway view of the main body, with a front portion removed, showing the upper cartridge bore, the internal cartridge seat, the lower stem bore, the external thread, and the internal locking collar.

FIG. 6 is a perspective view of the stem, showing the shaft, the base flange, the locking lug, and the orientation marker notch.

FIG. 7 is a perspective cutaway view of the cartridge, showing the shell, the upper and lower frangible seals, and the vasoconstricting liquid.

FIG. 8 is a perspective view of the nose head, showing the internally threaded skirt, the mid-section, the domed upper end, and the exit hole.

FIG. 9 is a perspective cutaway view of the device in the primed (locked) state, with the cartridge and cotton element loaded and the stem locked.

FIG. 10 is a perspective cutaway view of the device in the deployed state, with the stem driven fully upward and the cotton element ejected through the nose head.

FIG. 11 is a perspective view of the main body.

FIG. 12 is a perspective view of the main body as seen from below.

FIG. 13 is a view of the main body from above, showing the upper opening and the external thread.

FIG. 14 is a view of the lower face of the main body, showing the lower opening, the keyed receiving feature, and the directional markings.

FIG. 15 is a perspective view of the main body showing the path traveled by the locking lug through the keyway.

FIG. 16 is a perspective sectioned view of the main body with the stem at the insertion position.

FIG. 17 is a perspective sectioned view of the main body with the stem at the locked position.

FIG. 18 is a perspective sectioned view of the main body with the stem at the deployment position.

FIG. 19 is a perspective view of the stem.

FIG. 20 is a perspective view of the stem showing the locking lug.

FIG. 21 is a front elevation view of the stem.

FIG. 22 is a perspective view of the cartridge.

FIG. 23 is a side elevation view of the cartridge.

FIG. 24 is a perspective view of the nose head.

FIG. 25 is a front elevation view of the nose head.

FIG. 26 is a sectioned front elevation view of the nose head.

FIG. 27 is a sectioned perspective view of the nose head.

DETAILED DESCRIPTION OF THE INVENTION

The following description sets forth one embodiment of the invention in detail, with dimensions given as approximate values for that embodiment. The invention is not limited to the specific dimensions, materials, or configurations described. All materials named herein, including the polylactic acid (PLA) used to form the prototype components, are provided as examples only and may be substituted with any other suitable material without departing from the invention, and variations are described further below.

Overview of components

Referring to the figures, the device 100 comprises a main body 10, a stem 20, a cartridge 30, an absorbent cotton element 40, and a nose head 50. The main body 10 further carries directional markings 25 on its lower face, described below.

Main body

The main body 10 is an elongated tubular structure having an upper end 11 and a lower end 12. In the described embodiment the main body has an overall height of approximately 82.4 mm, comprising a main tube portion of approximately 75 mm in height atop a base flange 13 of approximately 5 mm in height, with an externally threaded portion extending approximately 2.4 mm above the main tube.

The main tube portion has an outer radius of approximately 9 mm along most of its length. At the lower end 12, the body widens into the base flange 13, a finger-grip region having an outer radius of approximately 15 mm, which allows the user to brace two fingers during actuation in the manner of a syringe flange.

The main body 10 is formed of a rigid polymer, in the described embodiment polylactic acid (PLA), for example the "PLA Basic" filament used in fused-deposition 3D printing of the

prototype. This material is exemplary only; the main body may be formed of any suitable rigid material, as described under Alternative Embodiments below, and the choice of material is not limiting.

The body defines an internal bore extending along its full length, the bore having two principal regions of differing diameter:

Upper cartridge bore. The upper portion of the bore opens at the upper end 11 through an upper opening 14 having a radius of approximately 6.5 mm, surrounded by a relatively thin wall. A short distance below the upper opening, the bore narrows at an internal cartridge seat 16, formed in the described embodiment by a curved transition of approximately 3.7 mm in length and a side wall portion of approximately 1.5 mm in length. A cartridge 30 dropped into the upper opening 14 descends until it rests on the seat 16 and cannot pass further downward.

Lower stem bore. The lower portion of the bore opens at the lower end 12 through a lower opening 15 having a radius of approximately 5 mm, sized to closely receive both the stem 20 and the cotton element 40. The cotton element 40 is loaded through the lower opening 15 and pushed upward until its lower end is substantially flush with the lower end 12 of the body.

External thread. The upper end 11 of the main body carries an external thread 18, in the described embodiment an ISO metric M16 x 1.5 right-hand thread (class 6g) of approximately 3 mm thread length, for attachment of the nose head 50.

Directional markings. The lower face of the base flange 13 carries directional markings 25 that indicate to the user which way to rotate the stem 20. In the described embodiment the markings comprise the word "LOCK" positioned on one side of the lower opening 15 with an adjacent arrow, and the word "PUSH" positioned on the opposite side of the lower opening 15 with an adjacent arrow. As viewed from below, that is, looking directly at the lower face of the base flange 13, the arrow associated with "LOCK" points to the right and indicates clockwise rotation, being the direction of rotation that seats the lug 23 in the locking pocket 19d for priming and storage; and the arrow associated with "PUSH" points to the left and indicates counterclockwise rotation, being the direction of rotation that releases the lug 23 from the locking pocket 19d and aligns it with the deployment channel 19e for actuation. The markings are arranged circumferentially about the lower opening 15, with each word and its arrow read together, and the words oriented radially about the lower opening.

The markings may be formed integrally with the main body 10, for example as raised or recessed lettering molded into the lower face of the base flange 13, or may alternatively be applied by printing, laser marking, engraving, pad printing, or an adhesive label. Forming the markings as integral raised or recessed relief is preferred in the described embodiment because the markings then remain legible over the life of the reusable device and cannot wear off through repeated handling.

The directional markings 25 cooperate with the orientation marker notch 24 on the stem 20 and the keyed receiving feature 17 of the main body: the notch 24 indicates the correct rotational orientation for inserting the stem, while the markings 25 indicate the correct direction to rotate the stem once inserted. Together these features allow a user to prime and deploy the device correctly without written instructions, which is significant because the device is intended for use during an active nosebleed, when a user may be under stress and unable to consult instructions.

Keyed stem interface and locking mechanism

The lower end 12 of the main body defines a keyed receiving feature 17 comprising a generally rectangular slot opening into the circular lower bore, such that the stem 20, which carries a corresponding radially projecting lug, can be inserted only in a specific rotational orientation, in the manner of a key entering a lock.

Within the lower bore, the body defines a longitudinal keyway bounded by two internal guide walls 19a and 19b disposed on generally opposite sides of the lug's path, in the manner of two parallel rails of differing lengths. In the described embodiment, the longer guide wall 19a extends approximately 60.4 mm and extends further downward than the shorter guide wall 19b, which extends approximately 55.4 mm. Near the lower end of the keyway, the body defines a transverse turning region 19c through which the lug of the stem may sweep circumferentially. In the described embodiment the turning region has an arc length of approximately 15.6 mm at a bore radius of approximately 5 mm, and a sweep height of approximately 5 mm. The longer guide wall 19a blocks the lug from crossing between channels along the length of the travel, while the turning region 19c permits the lug to sweep beneath the lower edge of the shorter guide wall 19b to pass between (a) a locking pocket 19d and (b) a deployment channel 19e.

The locking pocket 19d is a lower region of the keyway, in the described embodiment having wall heights of approximately 9 mm on its longer side and approximately 4 mm on its shorter side, dimensioned to receive the stem's lug in a close friction fit. When the lug is twisted into the locking pocket and the stem is pulled downward, the friction fit retains the stem securely in the primed position, such that the assembled device holds together solidly and does not come apart in storage. The friction retention requires that the stem first be pushed upward, relieving the friction engagement, before the stem can be twisted out of the locking pocket.

Stem

The stem 20 comprises an elongated cylindrical shaft 21 and a base flange 22. In the described embodiment the shaft 21 has a radius of approximately 4.85 mm and a height of approximately 73 mm, and the base flange 22 has a radius of approximately 15 mm and a height of approximately 5 mm, for an overall stem height of approximately 78 mm. The shaft radius of

approximately 4.85 mm provides a close sliding fit within the lower bore of radius approximately 5 mm.

The stem 20 is formed of a rigid polymer, in the described embodiment polylactic acid (PLA), for example "PLA Basic" filament. As with the other components, this material is exemplary and not limiting.

Locking lug. A radially projecting locking lug 23 is disposed on the shaft 21, its upper edge located approximately 8 mm below the top of the shaft. In the described embodiment the lug 23 is formed as the solid region between two circular arcs, one of radius approximately 4.85 mm (the shaft surface) and one of radius approximately 6.325 mm (the lug's outer extent), cut at an angle to produce four edge lengths of approximately 1.48 mm each, the lug having a height of approximately 4.8 mm. The lug 23 rides within the keyway of the main body, engages the locking pocket 19d in the primed state, and travels along the deployment channel 19e during actuation.

Orientation marker. The stem carries an orientation marker notch 24 at the front of its base region, in the described embodiment a square projection approximately 1 mm by 1 mm in cross-section extruded to a height of approximately 5 mm, which visually indicates to the user the correct rotational orientation for inserting the stem into the keyed receiving feature 17.

Cartridge

The cartridge 30 is a replaceable, generally cylindrical container that holds a vasoconstricting liquid 31. In the described embodiment the cartridge is an open-ended cylindrical shell having an inner radius of approximately 5.2 mm, an outer radius of approximately 6 mm (wall thickness approximately 0.8 mm), and a height of approximately 16 mm. The outer radius of approximately 6 mm allows the cartridge to drop into the upper cartridge bore of radius approximately 6.5 mm and seat on the cartridge seat 16.

The cartridge shell is formed of a thin plastic material, in the described embodiment the same polylactic acid (PLA) used to form the main body, for example "PLA Basic" filament. The cartridge is a disposable component that is replaced after use. This material is exemplary only and not limiting; other suitable shell materials are described under Alternative Embodiments below.

Both the upper and lower open ends of the cartridge shell are closed by frangible plastic seal membranes 32a (upper) and 32b (lower), which are sealed onto the shell and retain the liquid within the cartridge interior. In the described embodiment the cartridge interior is approximately three-quarters full of the vasoconstricting liquid 31, corresponding to approximately 1.0 mL of liquid (the full interior volume being approximately 1.36 mL).

In the described embodiment the vasoconstricting liquid 31 is a nasal decongestant solution whose active agent is oxymetazoline, a topical vasoconstrictor which constricts the blood

vessels of the nose (such solutions being commercially available, for example, under the brand Afrin).

Nose head

The nose head 50 is formed of a flexible, resilient material. In the described embodiment the nose head is formed of thermoplastic polyurethane (TPU), a flexible elastomeric material that provides a soft, atraumatic interface with the user's nostril. The material is exemplary only and not limiting; in alternative embodiments the nose head may be formed of other flexible biocompatible elastomers, including silicone rubber or other thermoplastic elastomers.

In the described embodiment the nose head 50 has an overall height of approximately 21.5 mm and comprises, from bottom to top: a lower internally threaded skirt section 51 of approximately 3.6 mm in height carrying an internal ISO metric M16 x 1.5 right-hand thread (class 6H) that mates with the external thread 18 of the main body; a cylindrical mid-section 52 of approximately 12 mm in height; and a domed upper end 53 of approximately 4.15 mm in height. The domed upper end 53 is formed as a 180-degree dome generated from a circle of radius approximately 9 mm, through which a central exit hole 54 of radius approximately 5 mm is cut. The widest portion of the nose head has a radius of approximately 9 mm, flush with the outer surface of the main body when installed, and the slimmer portion above the threaded skirt has a radius of approximately 6.5 mm. The internal passage through which the cotton element passes has a radius of approximately 5.75 mm.

The nose head 50 is screwed onto the upper end 11 of the main body, requiring several turns to seat flush against the body. When installed, the nose head 50 closes over the upper opening 14 and bears over the seated cartridge 30, retaining the cartridge in place between the cartridge seat 16 below and the nose head above. The rounded dome 53 permits comfortable insertion of the device's upper end into a nostril.

Absorbent cotton element

The absorbent cotton element 40 is a replaceable roll of absorbent cotton, substantially similar in construction to a dental cotton roll, dimensioned to be received through the lower opening 15 of the main body (radius approximately 5 mm) and to pass through the cartridge interior (inner radius approximately 5.2 mm) and the nose head exit passage. In the described embodiment the cotton element has a length of approximately 1.5 inches (approximately 38 mm) and a diameter of approximately 3/8 inch (approximately 9.5 mm), and is compressible so as to be received through and pass along the bores described. The cotton element is biodegradable. These dimensions are exemplary and not limiting.

The cotton element 40 serves the dual function of (a) absorbing the vasoconstricting liquid 31 as it passes through the cartridge interior during deployment, and (b) absorbing blood within the nostril after deployment.

Priming and storage

To prepare the device for use, the user:

1. Loads a cotton element 40 through the lower opening 15, pushing it upward until its lower end is flush with the lower end 12 of the body.
2. Drops a cartridge 30 into the upper opening 14, where it descends and rests on the cartridge seat 16.
3. Screws the nose head 50 onto the external thread 18 until flush with the body, thereby capturing the cartridge 30 in place.
4. Inserts the stem 20 into the keyed receiving feature 17 at the lower end, aligning the orientation marker notch 24, until the stem reaches its insertion stop a short distance in.
5. Twists the stem in the direction indicated by the "LOCK" marking on the lower face of the base flange 13, being clockwise as viewed from below, sweeping the lug 23 through the turning region 19c into the locking pocket 19d, and pulls the stem downward, seating the lug in the friction-fit locking pocket.

In this primed state the device is fully assembled, held solidly together by the friction engagement of the lug in the locking pocket, and ready for immediate use. The device is compact enough for storage in a first aid kit, nurse's office, sports bag, or other readily accessible location.

Deployment (operation during a nosebleed)

When a nosebleed occurs, the user:

1. Inserts the domed nose head 50 of the primed device into the affected nostril.
2. Pushes the stem 20 upward, relieving the friction engagement of the lug 23 in the locking pocket 19d.
3. Twists the stem fully in the direction indicated by the "PUSH" marking on the lower face of the base flange 13, being counterclockwise as viewed from below, sweeping the lug 23 through the turning region 19c, beneath the lower edge of the shorter guide wall 19b, and into the deployment channel 19e.
4. Pushes the stem upward through its full travel until it stops. As the stem advances, its upper end drives the cotton element 40 upward through the lower bore. The cotton element punches through the lower frangible seal 32b of the cartridge, passes through the cartridge interior where it absorbs the vasoconstricting liquid 31, punches through the upper frangible seal 32a, continues through the exit passage of the nose head 50, and exits through the exit hole 54 into the nostril.

Because the cartridge 30 is captured between the cartridge seat 16 and the screwed-on nose head 50, only the cotton element passes into the nose; the cartridge itself remains fixed in place within the body. The frangible seals 32a and 32b, being sealed onto the cartridge shell, break

open under the force of the advancing cotton element but remain attached to the cartridge and are not released into the nose.

The deployed cotton element, carrying the absorbed vasoconstricting agent, contacts the bleeding site, constricting the damaged blood vessel while simultaneously absorbing blood. In typical use the bleeding stops within a few minutes, and the user removes the cotton element after approximately five minutes.

Reloading and reuse

After use, the user unscrews the nose head 50, removes and discards the spent cartridge 30, discards the used cotton element 40, and repeats the priming sequence with a fresh cartridge and cotton element. The main body 10, stem 20, and nose head 50 are reused indefinitely. Because the only consumed components, the cartridge and the cotton element, are inexpensive, and because the cotton element is biodegradable, the device remains economical over time and generates substantially less waste than single-use nasal packing products.

ALTERNATIVE EMBODIMENTS

The invention is not limited to the specific embodiment described above. Variations within the scope of the invention include, without limitation:

Alternative vasoconstricting agents. The liquid 31 may comprise any suitable vasoconstricting or hemostatic agent, including without limitation oxymetazoline, phenylephrine, epinephrine solutions, tranexamic acid solutions, or saline carriers combined with any of the foregoing, in any suitable concentration.

Alternative absorbent elements. The absorbent element 40 may comprise cotton, rayon, cellulose sponge, expandable compressed sponge, gauze, or any other absorbent material, in cylindrical or other form, including forms with an attached retrieval string to aid removal from the nostril.

Alternative seal constructions. The frangible seals 32a, 32b may comprise plastic film, foil, wax-coated membranes, or laminated membranes, and may be scored or weakened to control the break pattern. Alternatively, the cartridge may include a single seal, a plug displaced rather than punctured by the cotton element, or a valve opened by the advancing cotton element.

Alternative locking mechanisms. The stem-retention mechanism may comprise, in place of or in addition to the described friction-fit twist lock, a bayonet pin-and-L-slot engagement, a partial-thread engagement, a detent and recess engagement, a snap-fit, or a spring-loaded catch.

Alternative attachment of the nose head. The nose head may attach by threads as described, or alternatively by bayonet mount, snap fit, press fit, or magnetic retention.

Alternative directional markings. The directional markings 25 may comprise words in any language, symbols, arrows alone, color coding, tactile relief readable by touch, or any combination thereof, and may be located on the lower face of the base flange 13, on the outer circumference of the base flange, on the stem base flange 22, or on any other surface visible to the user during operation. The wording may differ from that described, for example "OPEN" and "CLOSE," "UNLOCK" and "LOCK," or "READY" and "FIRE," and the markings may be supplemented by an index mark on the stem base flange 22 that aligns with the respective marking to confirm that the stem has been rotated fully into position.

Sizes. The device may be produced in multiple sizes, including a smaller pediatric size and a larger adult size, with correspondingly scaled bores, cartridges, and cotton elements.

Single-use embodiment. A fully disposable single-use version may be produced in which the body, stem, and nose head are also formed of low-cost biodegradable material and the device is discarded after one use.

Pre-loaded consumable sets. The cartridge and cotton element may be packaged together as a matched refill set, optionally as a single combined consumable unit in which the cotton element and cartridge are pre-assembled into a sleeve that drops into the body as one piece.

Materials. The rigid components may be formed of any suitable rigid polymer (including 3D-printed prototype materials or injection-molded medical-grade polymers), metal, or composite. The nose head may be formed of any soft biocompatible elastomer, including silicone rubber or thermoplastic elastomer.

Multiple actuation profiles. The stem stroke and bore lengths may be varied to control the depth of cotton deployment, and the turning region may be configured for left-hand locking rather than right-hand locking.

CLAIMS (INFORMAL)

While claims are not required in a provisional application, the following informal claims illustrate the scope of the invention:

1. A nasal hemostatic device comprising: an elongated tubular body having an upper bore and a lower bore; a replaceable cartridge containing a vasoconstricting liquid and sealed by at least one frangible seal, the cartridge seated within the upper bore; a replaceable absorbent element received in the lower bore; a nose head attached to the upper end of the body and retaining the cartridge in place; and a stem received through the lower end of the body, wherein advancing the stem drives the absorbent element through the frangible seal and through the interior of the cartridge, whereby the absorbent element

absorbs the vasoconstricting liquid and is deployed through the nose head into a user's nostril.

2. The device of claim 1, wherein the stem carries a radially projecting lug and the body defines an internal keyway having a locking pocket and a deployment channel separated by guide walls of differing lengths, such that the stem may be twisted in a first direction and drawn downward into a friction-retained locked position for storage, and twisted in a second direction to align with the deployment channel for actuation.

3. The device of claim 1, wherein the body, stem, and nose head are reusable, the cartridge is a replaceable disposable component, and the absorbent element is a replaceable biodegradable component.

4. The device of claim 1, wherein the cartridge is sealed at both its upper and lower ends by frangible membranes that remain attached to the cartridge after being broken by the advancing absorbent element.

5. The device of claim 1, wherein the nose head is formed of a flexible elastomeric material and includes a domed upper end shaped for atraumatic insertion into a nostril.

6. The device of claim 2, further comprising directional markings carried on a lower face of the body indicating the direction of rotation for locking the stem and the direction of rotation for releasing the stem, the markings being disposed on opposite sides of the lower opening and each comprising a word and an associated directional arrow.

7. The device of claim 6, wherein the directional markings are formed as raised or recessed relief integral with the body, and cooperate with an orientation marker on the stem indicating the correct rotational orientation for insertion of the stem into the body.